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MC190202781: WAJID ALI

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 26 of 30 (Start time: 11:00:08 AM, 12 July 2020) Total Marks: 1

An element $F_n\left(\frac{1}{2}\right) \in F_n(x) = \sum_{n=1}^{\infty} x^n,\ n \in \mathbb{N}, x \in \mathbb{R}$, is convergent to ------

Select the correct option

Reload Math Equations

- 0
- 1
- 1/2
- infinity

Click to Save Answer & Move to Next Question

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 25 of 30 (Start time: 10:59:27 AM, 12 July 2020)

Total Marks: 1

The uniform convergence is --------to transfer the properties of Continuity, Differentiability or Integrability to limit function.

Select the correct option

- strong enough
- not strong enough

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 24 of 30 (Start time: 10:58:31 AM, 12 July 2020)

Total Marks: 1

For the power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$, the expression: $\int_{\alpha}^{\beta} f(x) = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(\alpha^n - \beta^n)$ holds only if

Select the correct option

Reload Math Equations

- in the interior
- in the exterior
- at the boundary
- at the equidistant from the boundary

Click to Save Answer & Move to Next Question

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For the power series: $f(x) = \sum_{n=0}^{\infty} c_n x^n, x \in \mathbb{R}$, the $c_n = ----$.

$$\frac{f^{(n)}(1)}{n!}$$

$$\frac{f^{(n)}(0)}{n!}$$



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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 23 of 30 (Start time: 10:57:52 AM, 12 July 2020) Total Marks: 1

In the definition of uniform convergence the natural number N always depends on

Select the correct option

Reload Math Equations

- ε (epsilon)
- η
- β
- None of these

Click to Save Answer & Move to Next Question

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 21 of 30 (Start time: 10:55:38 AM, 12 July 2020)

Total Marks: 1

The power series $\sum_{n=0}^{\infty}(n!)x^n$ converges -------.

Select the correct option

Reload Math Equations

- $\forall x \in (-1, 1)$
- $\forall x \in \mathbb{R}$
- for $x = 0$ only
- for $x = 1$ only

Click to Save Answer & Move to Next Question

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 22 of 30 (Start time: 10:56:43 AM, 12 July 2020)

Total Marks: 1

The method of testing the uniform convergence of a series $\sum f_j$ of functions by definition is hard because - - - .

Select the correct option

Reload Math Equations

- checking continuity of each f_j is not an easy task
- finding the S_n (nth partial sum) is not always easy

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 20 of 30 (Start time: 10:54:35 AM, 12 July 2020)

Total Marks: 1

If a series of functions $\sum f_n$ converges uniformly on S, and for every $\varepsilon > 0$,

$\|f_n\|_S + \|f_{n+1}\|_S + \cdots + \|f_m\|_S < \varepsilon$, whenever $m, n \geq$

Select the correct option

Reload Math Equations

- \left\| {f_n} + {f_{n + 1}} + \cdots + {f_m} \right\|_S < \varepsilon
- \left\| {f_n} + {f_{n + 1}} + \cdots + {f_m} \right\|_S \geqslant \varepsilon

Click to Save Answer & Move to Next Question

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Apps

MC100201148: WAQAS AHMAD

MTH631:Grand Quiz

Question # 20 of 30 (Start time: 10:46:27 AM, 12 July 2020)

Convergence of the sequence $\{F_n\}$ is NOT uniform, if there exists $\varepsilon > 0$ such that to each integer N there correspond an integer $n > N$ and a point $x_n \in S$ for which we have;

Select the correct option

○	$\lvert F_n(x_n) - F(x_n)\rvert \leqslant \varepsilon$
○	$\lvert F_n(x_n) - F(x_n)\rvert \geqslant \varepsilon$

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 17 of 30 (Start time: 10:51:35 AM, 12 July 2020)

Total Marks: 1

The condition of absolute convergence (pointwise or uniform) is ------ than the usual convergence(pointwise or uniform).

Select the correct option

- weaker
- stronger



Click to Save Answer & Move to Next Question

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MC190202781: WAJID ALI

Time Left 87 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 18 of 30 (Start time: 10:52:17 AM, 12 July 2020)

Total Marks: 1

If a series converges absolutely uniformly on S, then

Select the correct option

- it converges uniformly on S
- it diverges
- it is differentiable on S
- None of these

Click to Save Answer & Move to Next Question

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Time Left 87 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 16 of 30 (Start time: 10:50:27 AM, 12 July 2020) Total Marks: 1

The linear term of Taylor' series of $f(x) = \sin x$ about $x = 0$ is ;

Select the correct option

Reload Math Equations

- 1
- -1
- -x
- x

Click to Save Answer & Move to Next Question

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Topological Properties of $\mathbb{R}^n$ can be described in terms of special class of subsets, called- - - - - - - - - in $\mathbb{R}^n$.

Select the correct option

- open sets
- closed sets
- neighborhoods
- all are equivalent



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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 15 of 30 (Start time: 10:49:27 AM, 12 July 2020)

Total Marks: 1

The term by term differentiation of the series

$$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty$$

Select the correct option

Reload Math Equations

- $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty.$
- $\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty.$
- $\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty$
- $\sum_{n=20}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty.$

Click to Save Answer & Move to Next Question

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If $\{F_n\}$ converges pointwise to F on S, then $\{F_n\}$ converges _____ to F on S.

Select the correct option

- uniform
- pointwise
- piecewise
- non of these



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For a power series, if the radius of convergence is $R = \infty$, then power series is said to:

Select the correct option

- converge nowhere (converge at single point)
- converge everwhere in the given domain



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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 12 of 30 (Start time: 10:45:08 AM, 12 July 2020)

Total Marks: 1

Which of the following can be taken as dominating series for the uniform convergence of $\sum_{n=1}^{\infty}\left(\frac{1}{x^2+n^2}\right)$?

Select the correct option

Reload Math Equations

- $\sum\left(\frac{1}{n}\right)$
- $\sum\left(\frac{1}{n^2}\right)$
- $\sum n$
- $\sum n^2$

Click to Save Answer & Move to Next Question

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 11 of 30 (Start time: 10:44:20 AM, 12 July 2020)

Total Marks: 1

The continuity of limit function of the series solution of certain class of Partial differential equations can be proved by proving just its ..... convergence.

Select the correct option

- Pointwise
- Uniform

Click to Save Answer & Move to Next Question

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MC190202781: WAJID ALI

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 7 of 30 (Start time: 10:39:14 AM, 12 July 2020) Total Marks: 1

For a power series, if the radius of convergence is $R = \infty$, then power series is said to:

Select the correct option

Reload Math Equations

- converge nowhere (converge at single point)
- converge everwhere in the given domain

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MC190202781: WAJID ALI

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 9 of 30 (Start time: 10:41:42 AM, 12 July 2020)

Total Marks: 1

For the series of functions $\sum_{k=1}^{\infty} f_k$ defined on domain D, *if* the sequence of partial sum S_n converges uniformly on domain D,

then the given series - - - - - - - - - converges uniformly on D.

Select the correct option

Reload Math Equations

- necessarily
- not necessarily

Click to Save Answer & Move to Next Question

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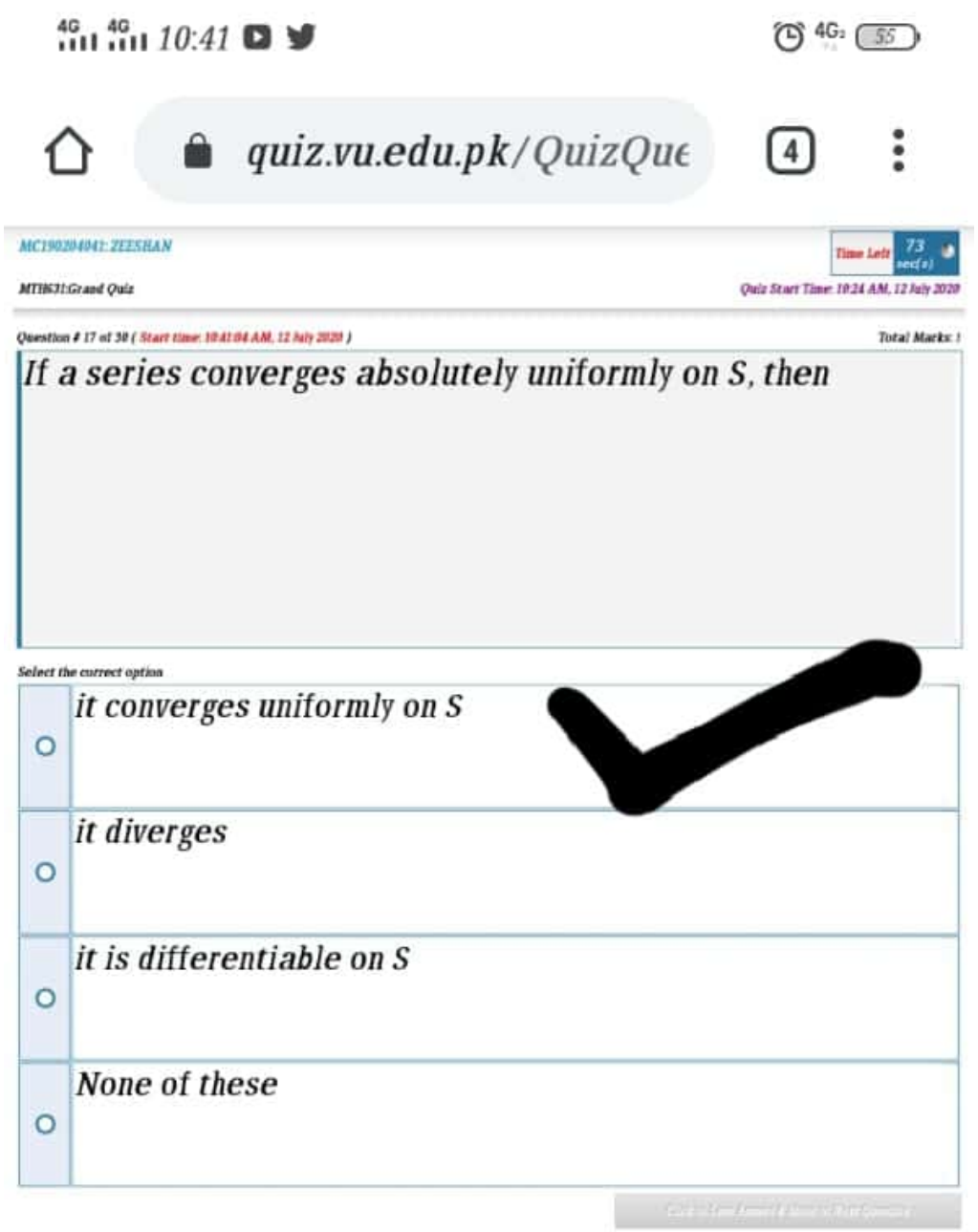
Question # 15 of 30 (Start time: 10:42:18 AM, 12 July 2020)

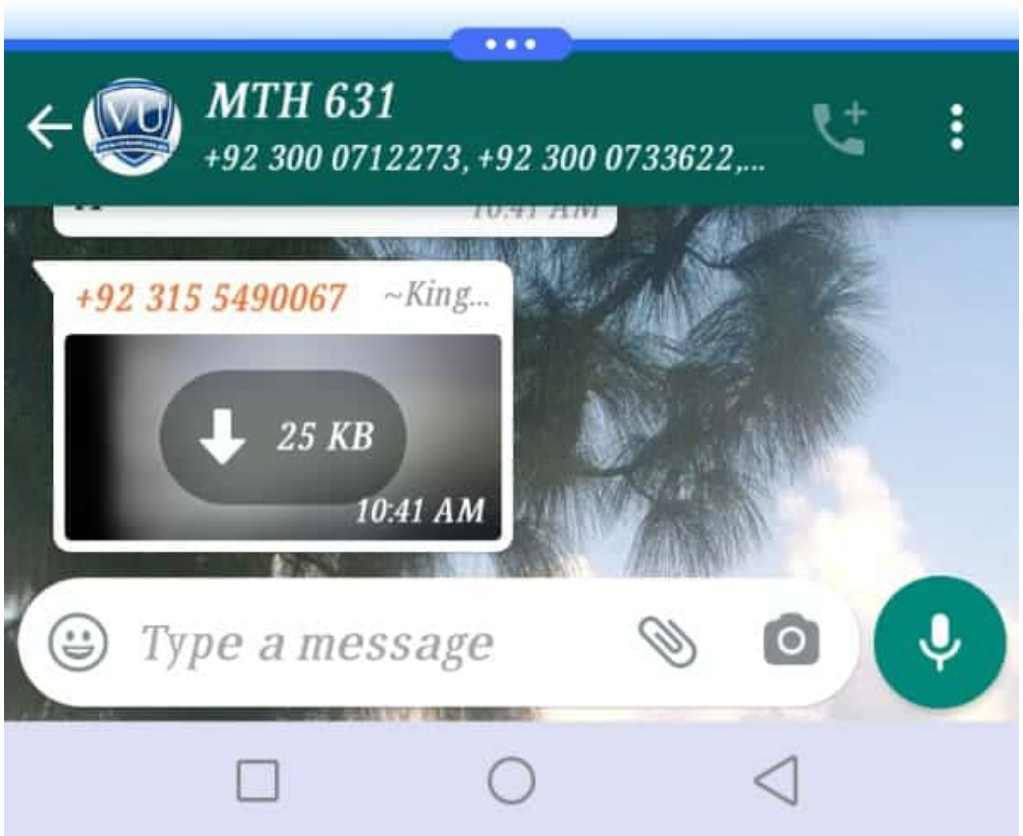
The limit function of the sequence of functions defined by $F_n(x) = x^n$, $x \in [0, 1]$ is - - - - - - .

Select the correct option

- Continuous
- Discontinuous

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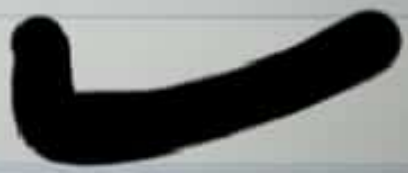
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Question # 14 of 30 (Start time: 10:40:57 AM, 12 July 2020)

If a series converges absolutely uniformly on S, then

Select the correct option:

- it converges uniformly on S
- it diverges
- it is differentiable on S
- None of these



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MTH631:Grand Quiz

Question # 14 of 30 (Start time: 10:38:35 AM, 12 July 2020)

Let $\{F_n(x)\}$ be a sequence of functions on $D \subset \mathbb{R}$, if to each $x \in D$ and to each $\varepsilon > 0$ there corresponds a positive integer m such that $|F_n(x) - F(x)| < \varepsilon \ \forall n \geq m$, then we say that $\{F_n(x)\}$ converges ----------, to the function $F(x)$ on D.

Select the correct option

- Point-wise
- Uniformly

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MC190201148: WAQAS AHMAD

MTH631:Grand Quiz

Question # 12 of 30 (Start time: 10:36:03 AM, 12 July 2020)

The pointwise convergence is ——to transfer the properties of Continuity, Differentiability or Integrability to limit function.

Select the correct option

- strong enough
- not strong enough

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MTH631:Grand Quiz

Question # 9 of 30 (Start time: 10:32:48 AM, 12 July 2020)

If a series of functions converges absolutely uniformly on a set S, then it ——— converge(s) uniformly on S.

Select the correct option

- ○ does not necessarily
- ○ necessarily



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Question # 11 of 30 (Start time: 10:34:47 AM, 12 July 2020)

If a series of functions converges uniformly on a set S, then it —— converge(s) absolutely uniformly on S

Select the correct option

- does not necessarily
- necessarily

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 4 of 30 (Start time: 10:35:12 AM, 12 July 2020) Total Marks: 1

If $f(x)$ is infinitely differentiable on $(-1,1)$, then Taylor series of $f(x)$ about $x=0$, can be expressed as;

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n, \text{ provided that; - - - - - - , where } a_n \in (-1,1).$$

Select the correct option

Reload Math Equations

- $\lim_{n\to\infty}\frac{f^{(n+1)}(a_n)}{(n+1)!}x^{n+1}<0$
- $\lim_{n\to\infty}\frac{f^{(n+1)}(a_n)}{(n+1)!}x^{n+1}>0$
- $\lim_{n\to\infty}\frac{f^{(n+1)}(a_n)}{(n+1)!}x^{n+1}=0$
- $\lim_{n\to\infty}\frac{f^{(n+1)}(a_n)}{(n+1)!}x^{n+1}=1$

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 3 of 30 (Start time: 10:33:44 AM, 12 July 2020) Total Marks: 1

A sequence of functions $\{F_n\}$ _____ on a set S if and only if for each $\varepsilon > 0$ there is an integer N such that $\|F_n - F_m\|_S < \varepsilon$ if $n, m \geq N$.

Select the correct option

Reload Math Equations

- converges uniformly
- diverges uniformly
- both a and b
- non of these

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 1 of 30 (Start time: 10:31:49 AM, 12 July 2020)

Total Marks: 1

For the power series; $f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$, $f'(x) = - - - -$.

Select the correct option

Reload Math Equations

- $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$
- $\sum_{n=0}^{\infty} (-1)^{n+1} \frac{x^{2n+1}}{(2n+1)!}$

Click to Save Answer & Move to Next Question

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MTH631:Grand Quiz

Quiz Start Time: 10:31 AM, 12 July 2020

Question # 2 of 30 (Start time: 10:32:58 AM, 12 July 2020)

Total Marks: 1

Which of the followings is not an axiom of the vector space V. $u, v, w \in V$ and $c \in F$.

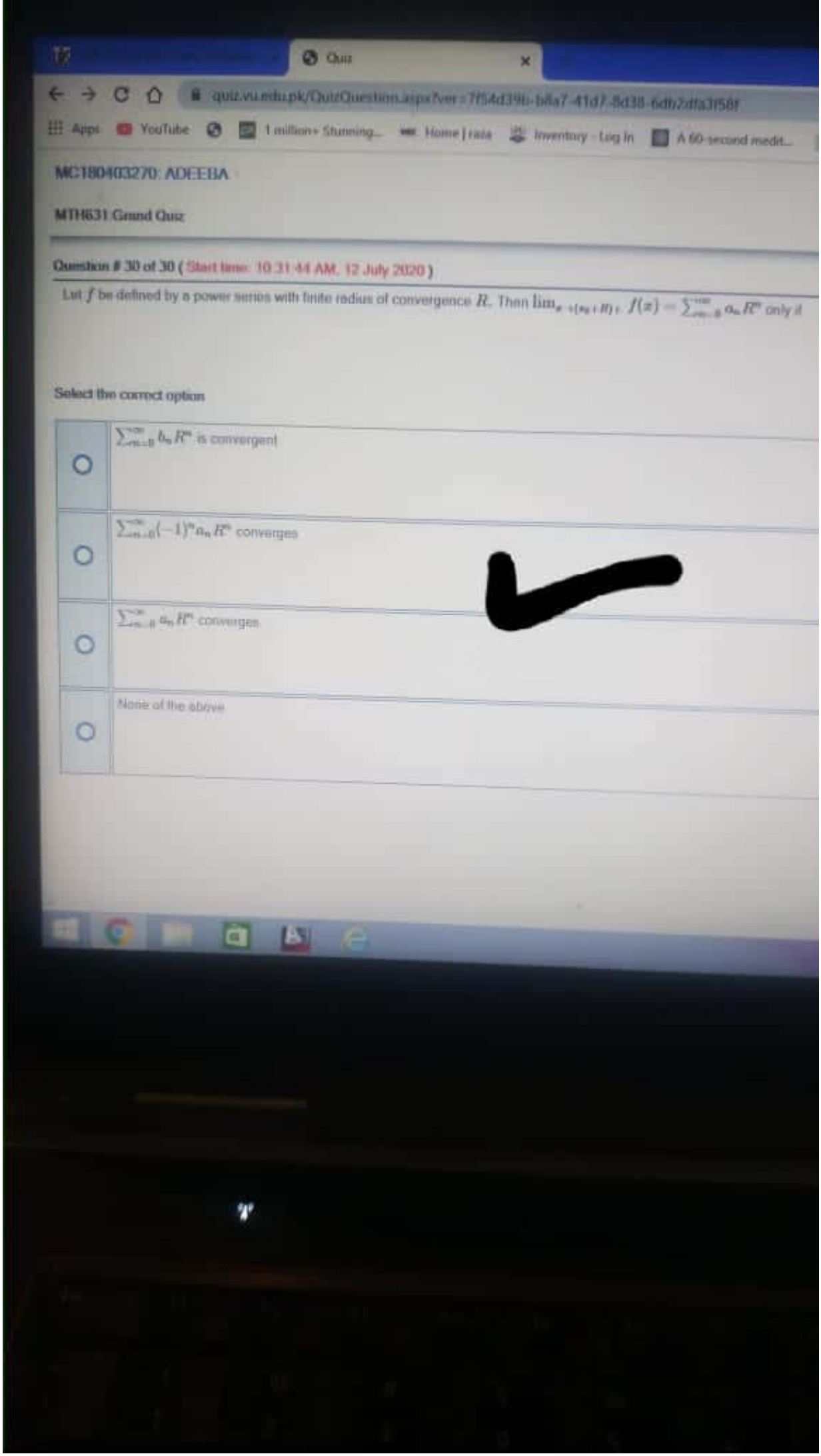
Select the correct option

Reload Math Equations

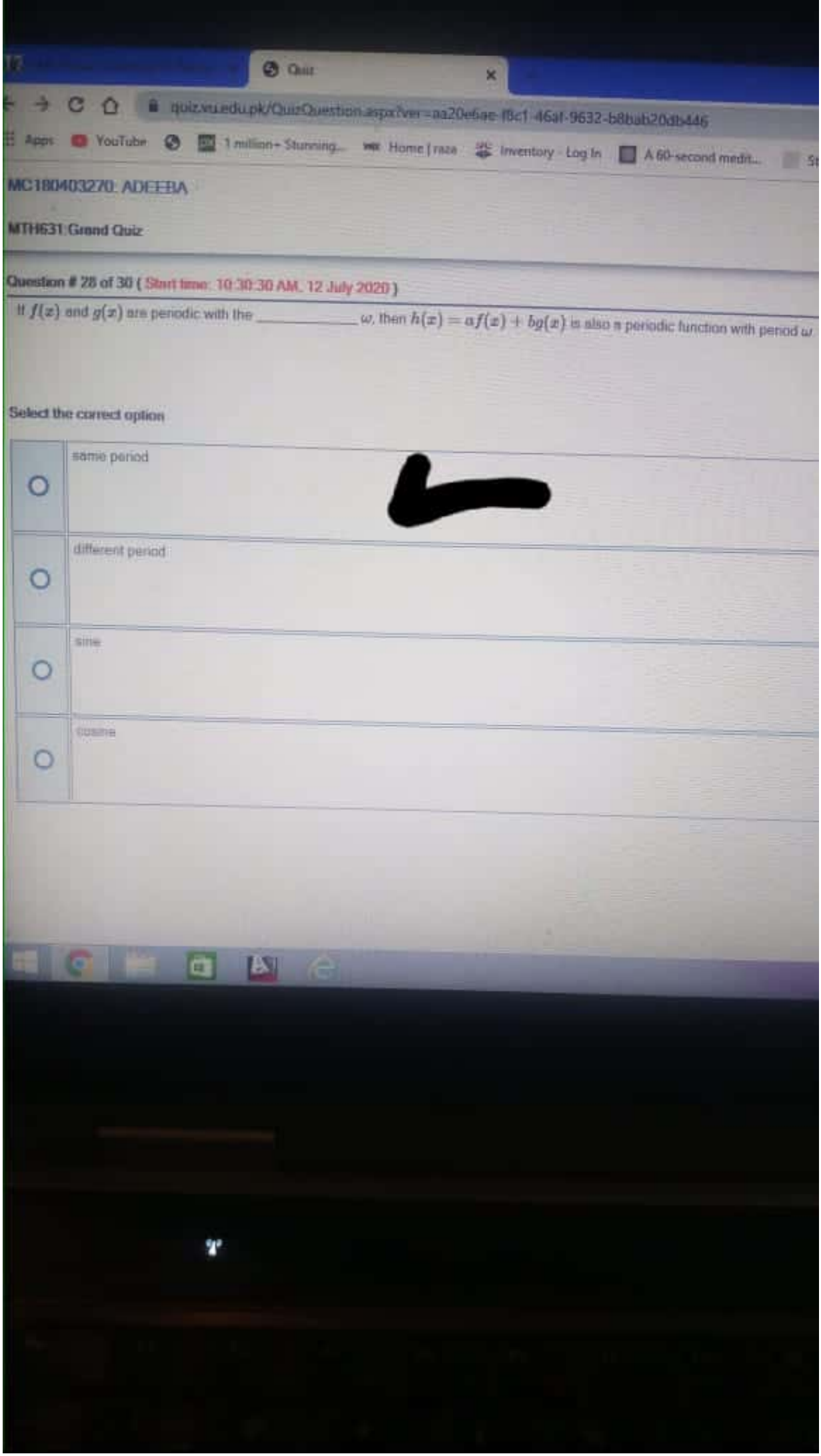
- $u + v = v + u$
- $u + (-u) = v$
- $c(u + v) = cu + cv$
- $(c + d)u = cu + du$

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 29 of 30 (Start time: 10:31:07 AM, 12 July 2020)

Which of the followings does not hold for an inner product function $\langle .,. \rangle : V \times V \to F$.

Select the correct option

- $\langle v, u \rangle = \langle u, v \rangle$
- $\langle u + v, w \rangle = \langle v, w \rangle + \langle u, w \rangle$
- $\langle cu, v \rangle = c \langle u, cv \rangle$
- Non of these

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 27 of 30 (Start time: 10:29:45 AM, 12 July 2020)

To check uniform convergence of $\sum \frac{1}{x^2+n^2}$ we can use and apply _____. Choose the best answer.

Select the correct option

- Weierstrass M-test
- dominated convergence test
- both a and b
- non of these

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 25 of 30 (Start time: 10:28:28 AM, 12 July 2020)

Fourier series expansion of a function is a representation of the function as a linear combination of

Select the correct option

- sines and tangents.
- cosines and tangents.
- tangents and cotangents.
- sines and cosines.

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For $x \neq 1$, the nth partial sum S_n of the series:
$1 + x + x^2 + \cdots + x^{n-1}$ as $n \to \infty$, is - - - - .

$\frac{1}{1+x}$



$\frac{1}{1-x}$

$\frac{1}{1+2^n}$

$\frac{1}{1-x^n}$

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 24 of 30 (Start time: 10:27:12 AM, 12 July 2020)

The power series expansion of $f(x) = \sin x$ about $x_0 = 0$ is

Select the correct option

○	$\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,$
○	$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty.$
○	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty,$
○	$\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty$

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Every power series $\sum_{n=0}^{\infty} a_n x^n$ with radius of convergence $R > 0$, is a Taylor's series about 0 of its sum function as $f(x) =$ .....

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{(n+1)!} x^n$$

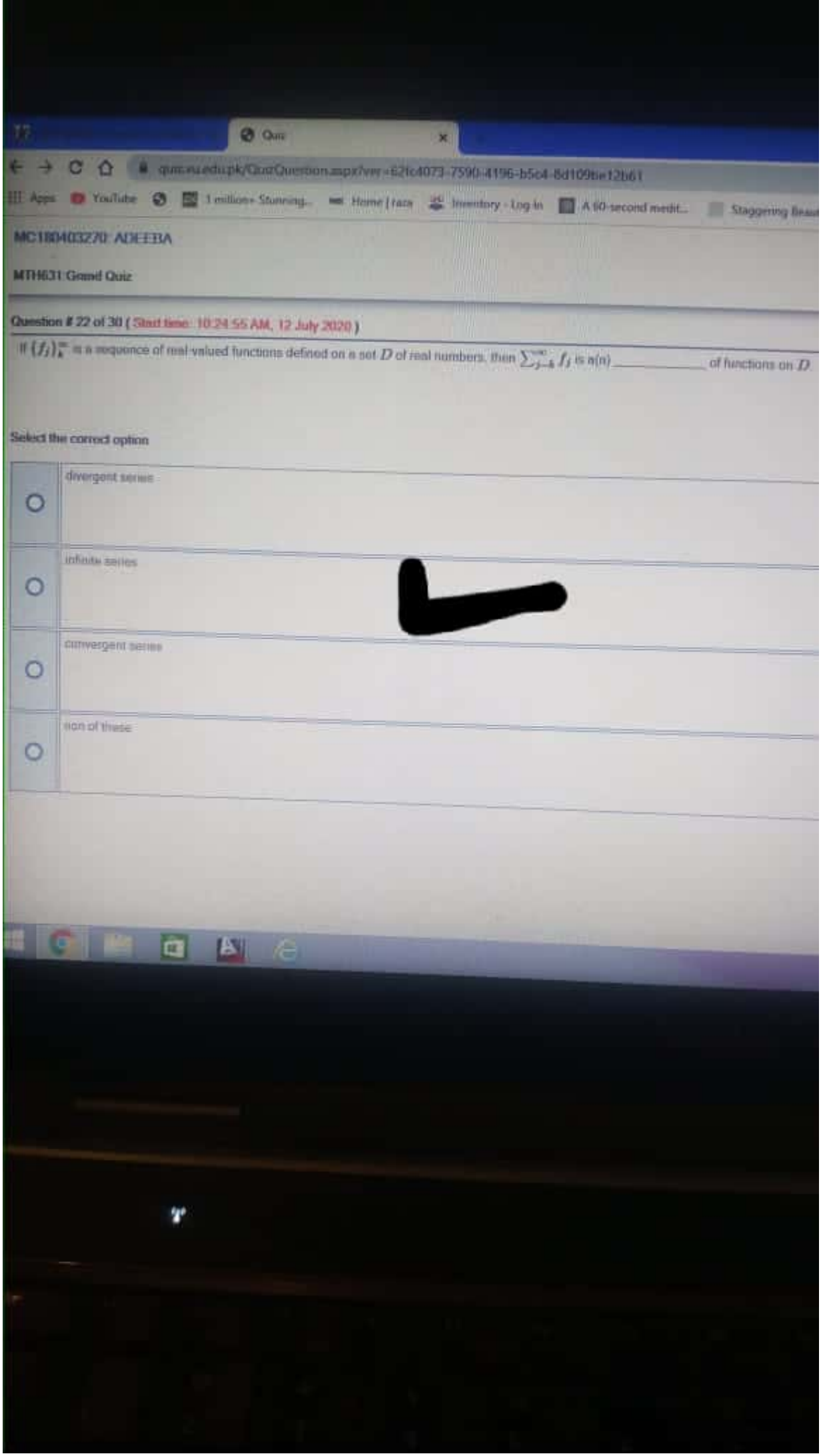
$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{(n-1)!} x^n$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

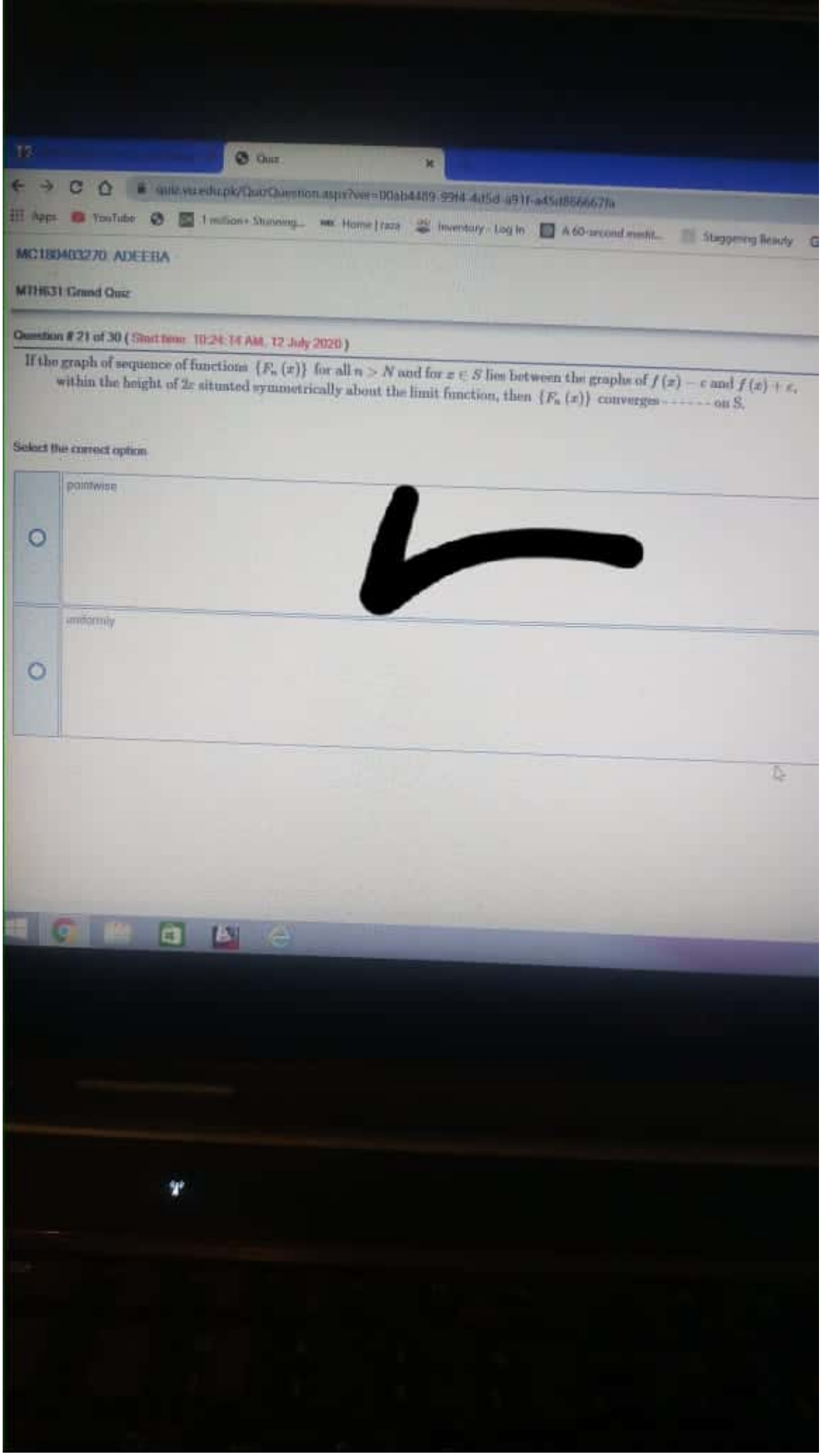


$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{(2n)!} x^n$$

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 19 of 30 (Start time: 10:21:25 AM, 12 July 2020)

If a power series $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ converges to a function $f(x)$, then the domain of $f(x)$ is ------ as that of the interval of convergence of power series.

Select the correct option

○	not necessarily the same
○	same and equal

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 18 of 30 (Start time: 10:20:14 AM, 12 July 2020)

By omitting or inserting some finite terms in a series will NOT affect which of the following?

Select the correct option

- Convergence or divergence of the series
- Sum of the series



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Quiz

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 17 of 30 (Start time: 10:19:18 AM, 12 July 2020)

If $f(x)$ is a periodic function with period ω, then $\int_a^{a+\omega} f(x)dx =$ _____ whenever f is integrable on $[0, \omega]$

Select the correct option

- $\int_a^{\omega} f(x)dx$
- $\int_0^{a} f(x)dx$
- $\int_0^{\omega} f(x)dx$
- non of these

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Quiz

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 16 of 30 (Start time: 10:18:02 AM, 12 July 2020)

The term by term differentiation of the series is $S(x) = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}$

Select the correct option

○	$\sum_{n=0}^{\infty}(-1)\frac{x^{2n+1}}{(2n+1)!}$
○	$\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$
○	$\sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!}$
○	None of these

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Quiz

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MC180403270: ADEEBA

MTH631:Grand Quiz

Question # 15 of 30 (Start time: 10:17:25 AM, 12 July 2020)

If $\sum f_n$ converges uniformly on S, then $\lim_{n \to \infty} \|f_n\|_S =$ _____

Select the correct option

- n
- f_n
- 1
- 0

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MTH631:Grand Quiz

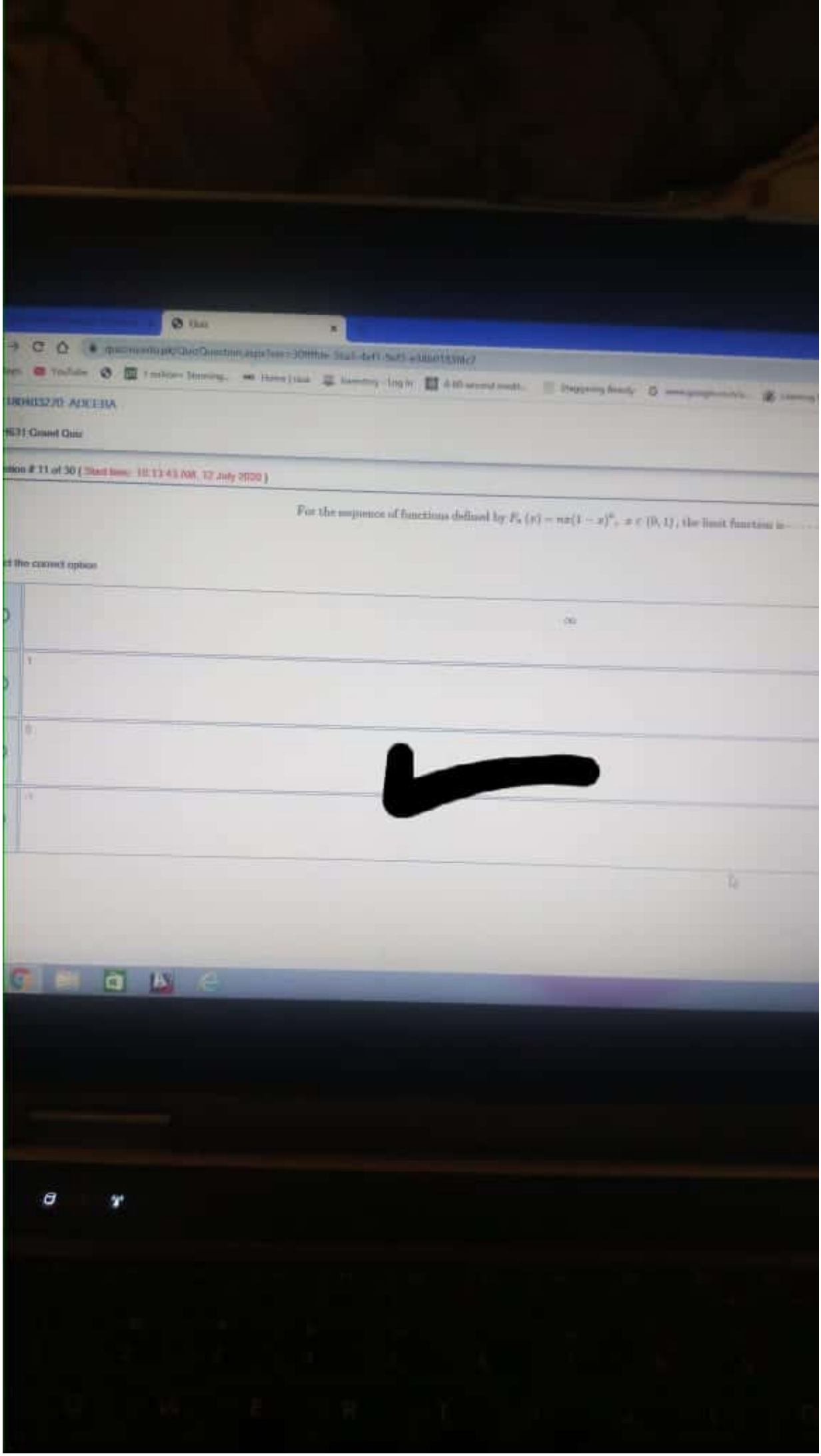
Question # 13 of 30 (Start time: 10:15:56 AM, 12 July 2020)

For the sequence of functions defined by $F_n(x) = \dfrac{x}{1+nx}$, $x \geqslant 0$,
the limit function $F(x) = \ldots$

Select the correct option

- 1
- 0
- n
- X

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Quiz

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MTH631:Grand Quiz

Question # 9 of 30 (Start time: 10:11:56 AM, 12 July 2020)

Topological Properties of R^n can be described in terms of special class of subsets, called---------- in R^n.

Select the correct option

- open sets
- closed sets
- neighborhoods
- all are equivalent

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$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad |x - x_0| < R,$$

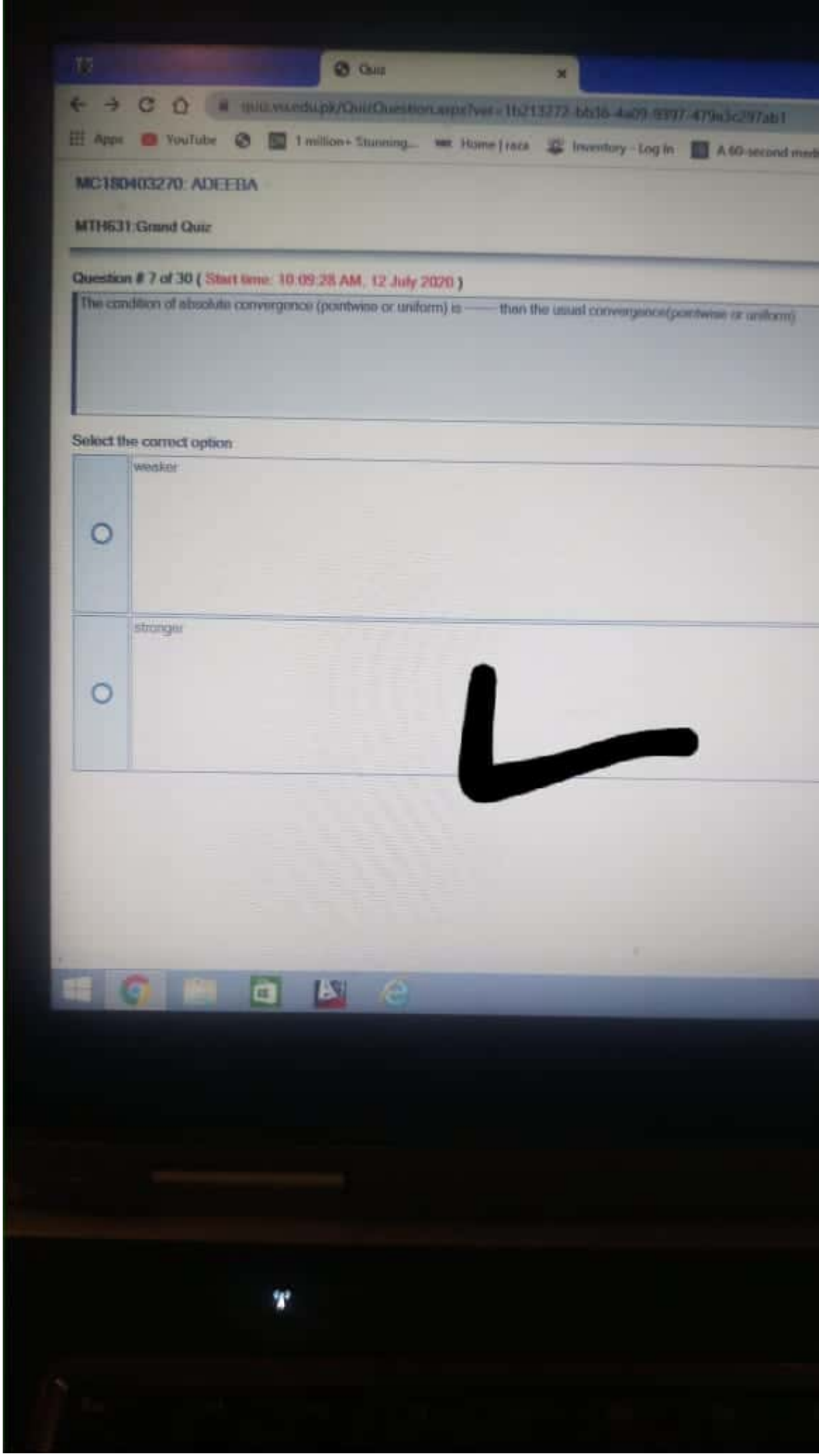
$$a_n = \frac{f^{(n)}(0)}{n!}.$$

$$a_n = \frac{f^{(2n)}(x_0)}{n!}.$$

$$a_n = \frac{f(nx_0)}{n!}.$$

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

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Question # 14 of 30 (Start time: 09:59:55 AM, 12 July 2020)

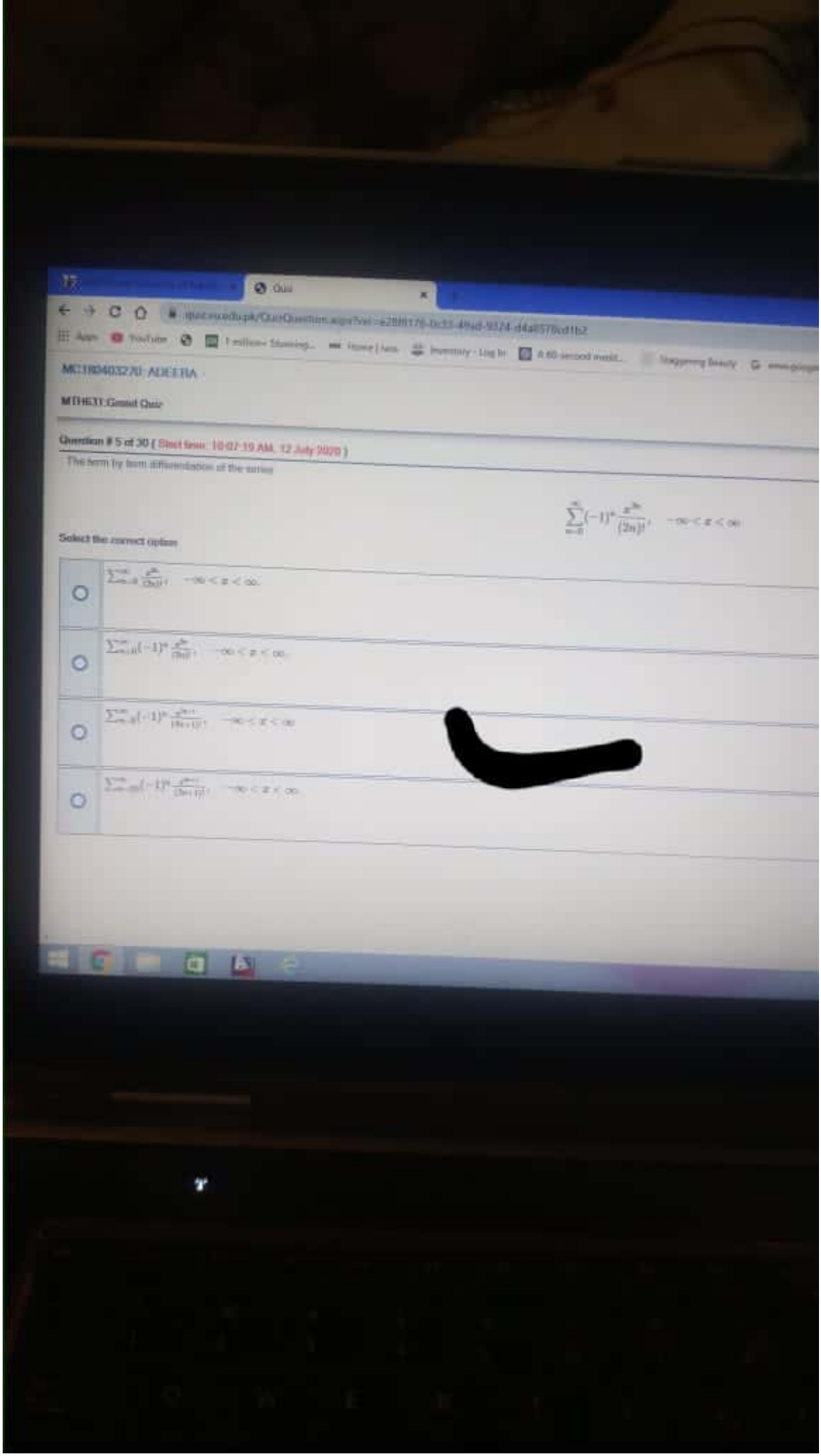
For the sequence of functions $\{F_n(x)\}$ defined and converges pointwise as;

$F_n(x) = \dfrac{nx}{1+n^2x^2}, x \in [0,1]$, the value of $\lim\limits_{n\to\infty} \|F_n(x) - F(x)\|_{[0,1]} \Rightarrow \{F_n(x)\}$ is $-----$, where $\lim\limits_{n\to\infty} F_n(x) = F(x)$.

Select the correct option

○	uniformly convergent
○	not uniformly convergent

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MC190200502: MUHAMMAD ANSAR

Time Left 88 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 09:08 AM, 12 July 2020

Question # 22 of 30 (Start time: 09:34:56 AM, 12 July 2020)

Total Marks: 1

For the power series: $f(x) = \sum_{n=0}^{\infty} c_n x^n, x \in \mathbb{R}$, the $c_n = - - - -$.

Select the correct option

Reload Math Equations

- $\frac{f^{(n)}(1)}{n!}$
- $\frac{f^{(n)}(0)}{n!}$



Click to Save Answer & Move to Next Question

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Quiz

MC180403270 ADEEBA

MTH631 Grand Quiz

Question # 4 of 30 (Start time: 10:06:11 AM, 12 July 2020)

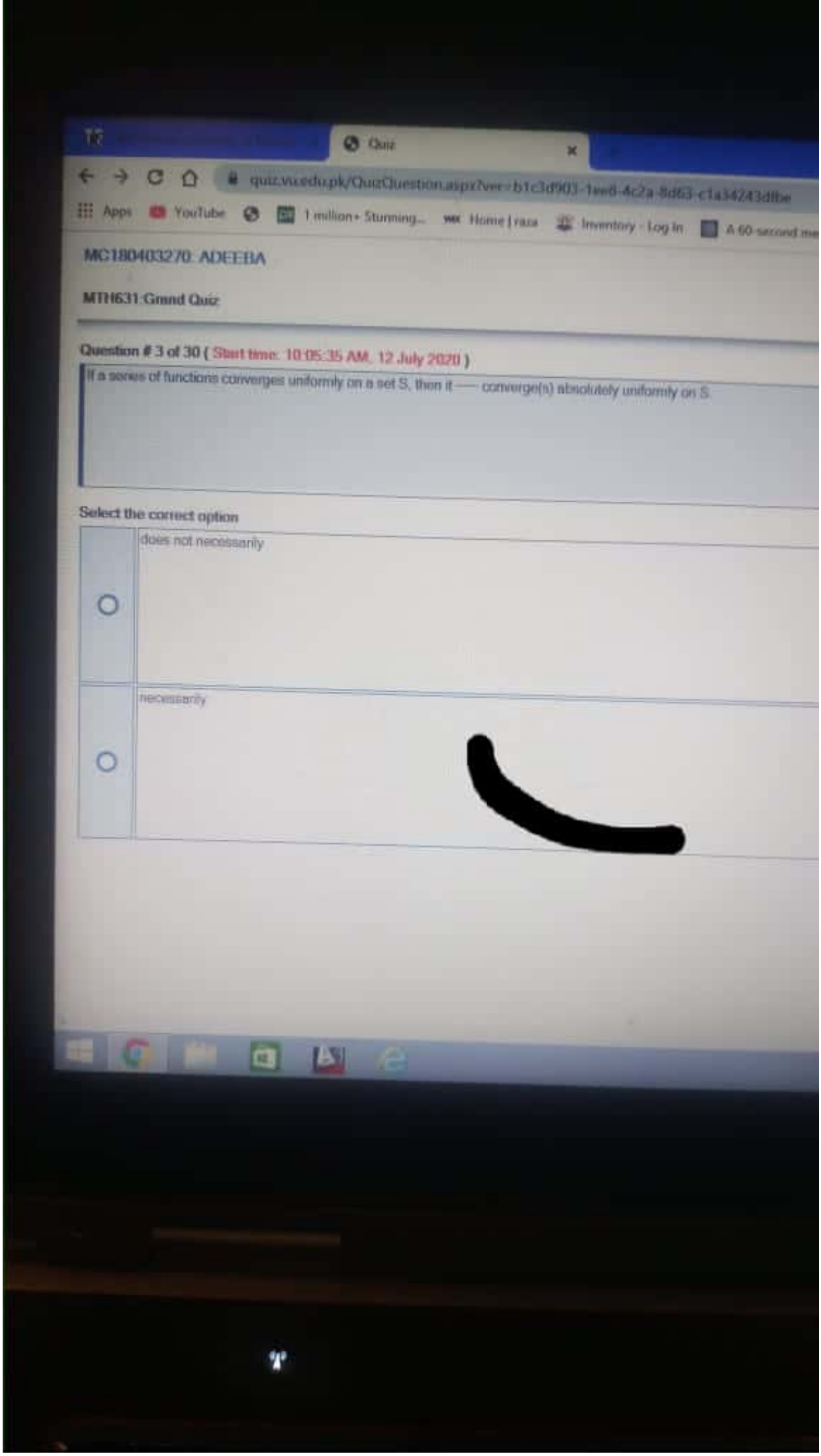
For the series of functions $\sum_{n=1}^{\infty} f_n$ defined on domain D, if the sequence of partial sum S_n converges uniformly on domain D, then the given series - - - - - - - - converges uniformly on D.

Select the correct option

necessarily

not necessarily

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Quiz

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MC180403270: ADEEBA

MTH631 Grand Quiz

Question # 2 of 30 (Start time: 10:04:48 AM, 12 July 2020)

Power series is a ---- case of series of functions in a given domain S.

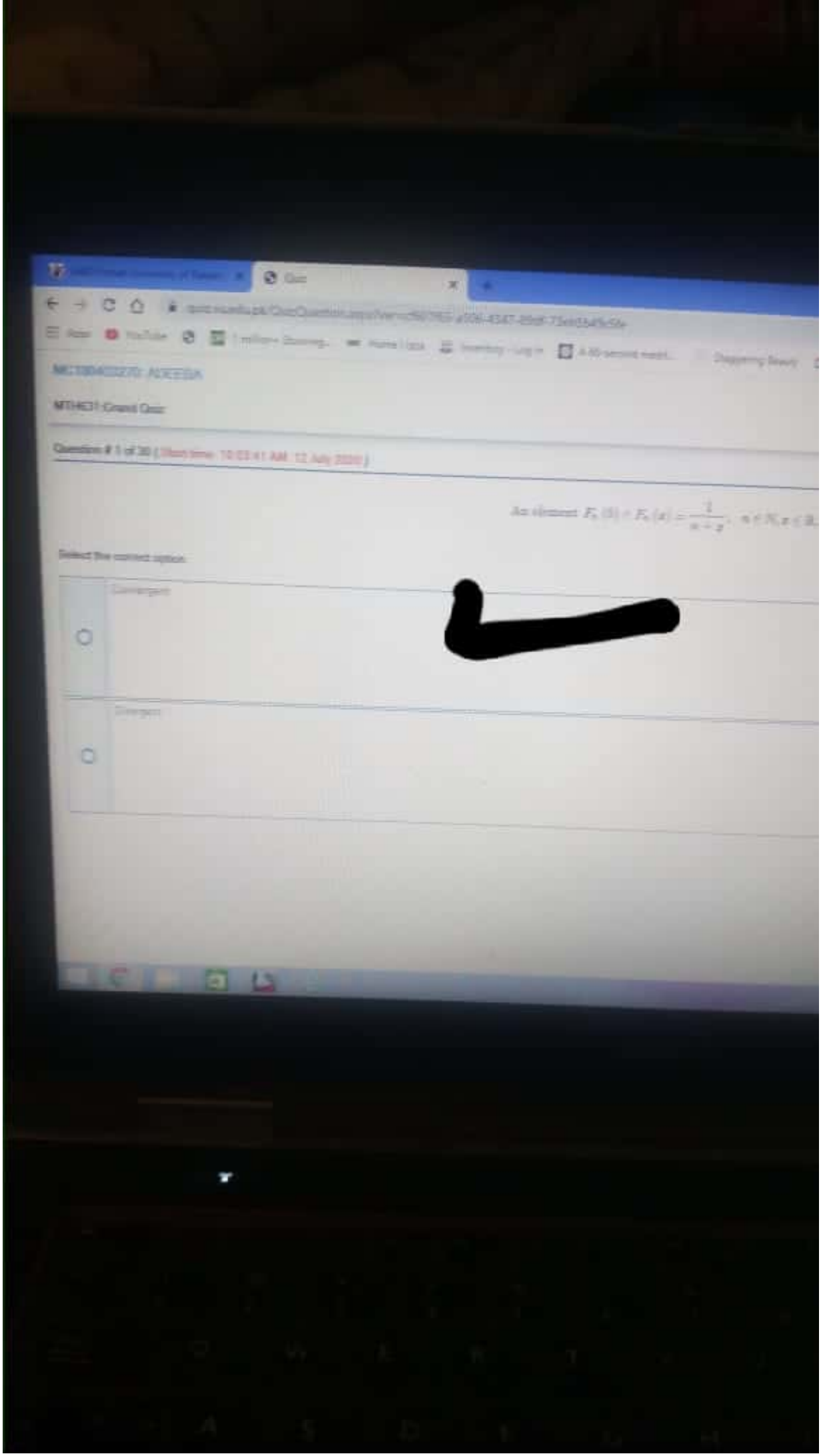
Select the correct option

general

special



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MTH431 Grand Quiz

Question # 1 of 30

Select the correct option

Convergent

Divergent

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MTH631 Grand Quiz

Quiz Start Time: 09:35 A

Question # 30 of 30 (Start time: 10:02:30 AM, 12 July 2020)

A power series can be differentiate | term by-term ——— the interval of convergence

Select the correct option

- on the exterior of
- in the interior of
- in the neighborhood
- at the extreme points of

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Quiz Start Time: 09:35 AM, 12 J

estion # 29 of 30 (Start time: 10:01:55 AM, 12 July 2020)

Total

Which of the following can be taken as dominating series for the uniform convergence of $\sum_{n=1}^{\infty}\left(\frac{1}{x^2+n^2}\right)$?

elect the correct option

Reload Math Equa

- $\sum\left(\frac{1}{n}\right)$
- $\sum\left(\frac{1}{n^2}\right)$
- $\sum n$
- $\sum n^2$

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Question # 28 of 30 (Start time 10:00:39 AM, 12 July 2020)

The series : $\sum_{n=1}^{\infty} (-1)^n$ - - - - -

Select the correct option

○	converges to 1
○	converges to -1
○	converges to 0
○	oscillating

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MTH631:Grand Quiz

Quiz Start Time

Question # 27 of 30 (Start time: 10:00:00 AM, 12 July 2020)

The values of x in a specific interval for which a given power series converges is called

Select the correct option

- Interval of continuity
- Interval of convergence
- radius of convergence
- None of these

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MTH631:Grand Quiz

Quiz Star

Question # 26 of 30 (Start time: 09:59:15 AM, 12 July 2020)

For the sequence of functions $\{F_n(x)\}$ defined and converges pointwise as;

$$\lim_{n\to\infty} F_n(x) = \frac{\sin nx}{\sqrt{n}} = 0 = F(x), x \in [0, \pi],$$

the value of $\lim \|F_n(x) - F(x)\|_{[0,\pi]} \Rightarrow \{F_n(x)\}$ is ----.

Select the correct option

- uniformly convergent
- non-uniformly convergent

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MTH631:Grand Quiz

Question # 22 of 30 (Start time: 09:55:18 AM, 12 July 2020)

If X and Y are in $\mathbb{R}^n$, then $|X+Y|^2 =$ ------.

Select the correct option

- $|X|^2 + 2|X||Y| + |Y|$
- $(|X| + |Y|)^2$
- $|X|^2 + 2(X.Y) + |Y|$
- $X^2 + 2|X||Y| + Y^2$

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Question # 23 of 30 (Start time 09:55:56 AM, 12 July 2020)

If X and Y are in $\mathbb{R}^n$, then $|X+Y|^2 \leqslant$ ------.

Select the correct option

○	$\|X\|^2 + 2(X.Y) + \|Y\|^2$
○	$\|\|X\|^2 - \|Y\|^2\|$
○	$\|\|X\| - \|Y\|\|^2$
○	$\|X\|^2 + 2\|X\|\|Y\| + \|Y\|^2$

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MTH631:Grand Quiz

Quiz Start Time: 09

Question # 17 of 30 (Start time: 09:49:12 AM, 12 July 2020)

To check uniform convergence of $\sum \frac{1}{x^2+n^2}$ we can use and apply _____ Choose the best answer

Select the correct option

Reload

- Weierstrass M-test
- dominated convergence test
- both a and b
- non of these



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Quiz Start Time: 09:35 AM

Question # 13 of 30 (Start time: 09:45:54 AM, 12 July 2020)

Fourier series expansion of a function is a representation of the function as a linear combination of

Select the correct option

- sines and tangents
- cosines and tangents
- tangents and cotangents
- sines and cosines



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MTH631:Grand Quiz

Question # 15 of 30 (Start time: 09:47:57 AM, 12 July 2020)

If a series of functions converges absolutely uniformly on a set S, then it ——— converge(s) uniformly on S

Select the correct option

- does not necessarily
- necessarily



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Quiz Start Time: 09:

Question # 14 of 30 (Start time 09:46:32 AM, 12 July 2020)

A sequence of functions $\{F_n\}$ _____ on a set S if and only if for each $\varepsilon > 0$ there is an integer N such that $\|F_n - F_m\|_S < \varepsilon$ if $n, m \geq N$

Select the correct option

Reload

- converges uniformly
- diverges uniformly
- both a and b
- non of these

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MTH631 Grand Quiz

Question # 11 of 30 (Start time: 09:44:06 AM, 12 July 2020)

If a power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges to a function $f(x)$, then the domain of $f(x)$ is ------ as that of the interval of convergence of power series.

Select the correct option

- not necessarily the same
- same and equal

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MTH631:Grand Quiz

Quiz Start T

Question # 12 of 30 (Start time: 09:45:23 AM, 12 July 2020)

For the sequence of function: defined by $F_n(x) = \frac{\sin nx}{\sqrt{n}}, x \in \mathbb{R}$

the identity: $\frac{d}{dx}\left(\lim_{n\to\infty} F_n(x)\right) = \lim_{n\to\infty}\left(\frac{d}{dx} F_n(x)\right)$ is------.

Select the correct option

- True
- False



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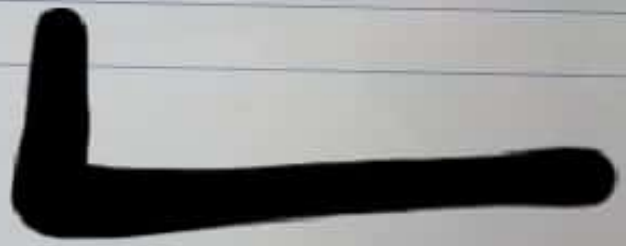
Quiz Start Time: 09:35 A

Question # 10 of 30 (Start time: 09:43:39 AM, 12 July 2020)

Point-wise convergence for the sequence of functions, is a ——property

Select the correct option

- local
- global



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MTH631 Grand Quiz

Question # 4 of 30 (Start time: 09:38:35 AM, 12 July 2020)

If $f(x)=\sum_{n=0}^{\infty} a_n(x-x_0)^n$, having $R>0$ as a radius of convergence, and $f(x)$ is continuous and differentiable in its interval of convergence, then radius of convergence of $f'(x)$ is - - - - - - .

Select the correct option

○	also R
○	$R+h, h>0$
○	$R-h, h<0$
○	$\frac{1}{R}, R \neq 0$

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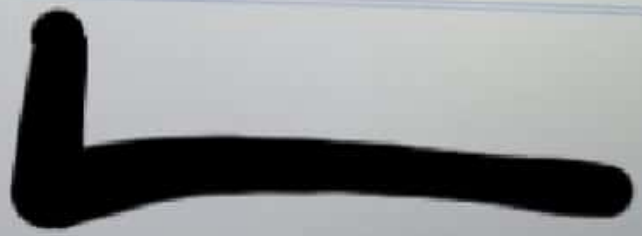
MTH631 Grand Quiz

Question # 23 of 30 (Start time: 09:28:59 AM, 12 July 2020)

A sequence of differentiable functions converging point-wise to a differentiable function, the limit of derivatives is ——equal the derivative of the limit function.

Select the correct option

- not
- necessarily



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Question # 3 of 30 (Start time: 09:37:30 AM, 12 July 2020)

If $f(x)$ and $g(x)$ are periodic with the _____ ω, then $h(x) = af(x) + bg(x)$ is also a periodic function with period ω.

Select the correct option

○	same period
○	different period
○	sine
○	cosine

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Question # 2 of 30 (Start time: 09:36:34 AM, 12 July 2020)

If $\{f_j\}_k^\infty$ is a sequence of real-valued functions defined on a set D of real numbers, then $\sum_{j=k}^{\infty} f_j$ is a(n) _____ of functions on D.

Select the correct option

- divergent series
- infinite series
- convergent series
- non of these

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MTH631:Grand Quiz

Question # 1 of 30 (Start time: 09:35:48 AM, 12 July 2020)

If the sequence of functions defined and converges pointwise as,

$$\lim_{n\to\infty} F_n(x) = \lim_{n\to\infty}\left(1-\frac{x_n}{n}\right) = 1 = F(x), 0 \leq x \leq 1.$$

Then the value of $\sup|F_n(x) - F(x)| \to |F_n(x)|$ converges

Select the correct option

- uniformly
- not uniformly

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MTH631:Grand Quiz

Question # 25 of 30 (Start time: 09:30:54 AM, 12 July 2020)

The sequence $\{F_n\}$ defined by $F_n = x^n, n \geq 1$ converges uniformly on every - - - - - - - subset of $[0,1)$.

Select the correct option

- bounded
- countable
- open
- closed

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MC190201711: JAFFAR HUSSAIN

MTH631:Grand Quiz

Question # 29 of 30 (Start time: 09:34:20 AM, 12 July 2020)

Which of the following can be served as counterexample for scalar multiplication operation in $\mathbb{R}^n$ for the axiom: $1X = X, \forall X \in \mathbb{R}^n$?

Select the correct option

- $\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n), \alpha \in \mathbb{R}$
- $\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, 0, \ldots, 0), \alpha \in \mathbb{R}$

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MTH631:Grand Quiz

Question # 30 of 30 (Start time: 09:34:45 AM, 12 July 2020)

An element $F_n(0) \in F_n(x) = \dfrac{1}{n+x}$, $n \in \mathbb{N}, x \in \mathbb{R}$, is ------.

Select the correct option

- Convergent
- Divergent

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MC190201711: JAFFAR HUSSAIN

MTH631:Grand Quiz

Question # 26 of 30 (Start time: 09:31:58 AM, 12 July 2020)

In the definition of uniform convergence the natural number N always depends on

Select the correct option

- ε (epsilon)
- η
- β
- None of these

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MC190200502: MUHAMMAD ANSAR

Time Left 88 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 09:08 AM, 12 July 2020

Question # 9 of 30 (Start time: 09:19:17 AM, 12 July 2020) Total Marks: 1

The Taylor series of an infinitely many differentiable function " f(x) " -------- converges to " f(x)".

Select the correct option

- not necessarily
- always

Click to Save Answer & Move to Next Question

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MC190200502: MUHAMMAD ANSAR

Time Left 89 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 09:08 AM, 12 July 2020

Question # 18 of 30 (Start time: 09:29:14 AM, 12 July 2020) **Total Marks:** 1

A power series can be differentiated term-by-term --------- the interval of convergence.

Select the correct option

- in the exterior of
- in the interior of
- in the neighborhood
- at the extreme points of

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Apps Gmail YouTube Maps

MC190200502: MUHAMMAD ANSAR

Time Left 88 sec(s)

MTH631:Grand Quiz

Quiz Start Time: 09:08 AM, 12 July 2020

Question # 16 of 30 (Start time: 09:27:23 AM, 12 July 2020) Total Marks: 1

If a series of functions converges uniformly on a set S, then it ----- converge(s) absolutely uniformly on S.

Select the correct option

- does not necessarily
- necessarily



Click to Save Answer & Move to Next Question

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MC180405712: SEHR AYUB

MTH631:Grand Quiz

Quiz Start Time: 08:31 AM, 12 Jul

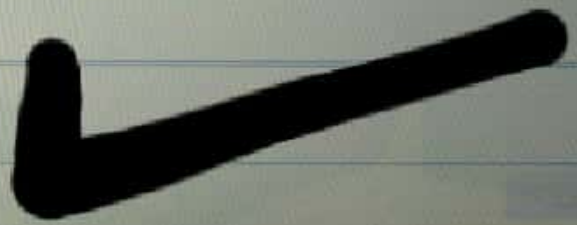
Question # 15 of 30 (Start time: 08:48:28 AM, 12 July 2020)

Total M

Fourier series expansion of a function is a representation of the function as a linear combination of

Select the correct option

- sines and tangents
- cosines and tangents
- tangents and cotangents
- sines and cosines



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Quiz

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MC190201221: SANIA

MTH631:Grand Quiz

Question # 9 of 30 (Start time: 08:41:55 AM, 12 July 2020)

Which of the followings is not an axiom of the vector space V. $u, v, w \in V$ and $c \in F$.

Select the correct option

- $u + v = v + u$
- $u + (-u) = v$
- $c(u + v) = cu + cv$
- $(c + d)u = cu + du$

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MC190201221: SANIA

MTH631:Grand Quiz

Question # 11 of 30 (Start time: 08:45:51 AM, 12 July 2020)

For the series of functions $\sum_{j=k}^{\infty} f_k$ defined on domain D, *if* the sequence of partial sum S_n converges uniformly on domain D, then the given series - - - - - - - - - converges uniformly on D.

Select the correct option

- necessarily
- not necessarily

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MC180405712: SEHR AYUB

MTH631:Grand Quiz

Quiz Start Time: 08:31 A

Question # 9 of 30 (Start time: 08:40:27 AM, 12 July 2020)

The limit of a uniformly convergent series of integrable functions is always

Select the correct option

Reload Math

- not integrable
- differentiable
- integrable
- None of these

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MC180405712: SEHR AYUB

MTH631 Grand Quiz

Question # 4 of 30 (Start time: 08:34:35 AM, 12 July 2020)

Quiz Start Time: 08:31 AM, 12 July 2020

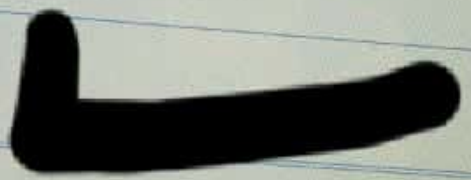
A sequence of functions $\{F_n\}$ _____ on a set S if and only if for each $\epsilon > 0$ there is an integer N such that $\|F_n - F_m\|_S < \epsilon$ if $n, m \geq N$

Total Marks: 1

Select the correct option

- converges uniformly
- diverges uniformly
- both a and b
- non of these

Reload Math Equations



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MC180405712: SEHR AYUB

MTH631 Grand Quiz

Question # 5 of 30 (Start time: 08:35:57 AM, 12 July 2020)

Time Left 86

Quiz Start Time: 08:31 AM, 12 July 2020

Total Marks: 1

If $f(x)$ and $g(x)$ are periodic with the _____ ω, then $h(x) = af(x) + bg(x)$ is also a periodic function with period ω.

Select the correct option

- same period
- different period
- sine
- cosine

Reload Math Equations

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MTH631:Grand Quiz

Question # 3 of 30 (Start time: 08:33:47 AM, 12 July 2020)

A sequence of differentiable functions converging uniformly to a differentiable function, the limit of derivatives is ---- equal the derivative of the limit function

Select the correct option

- not
- necessarily



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MC190201221: SANIA

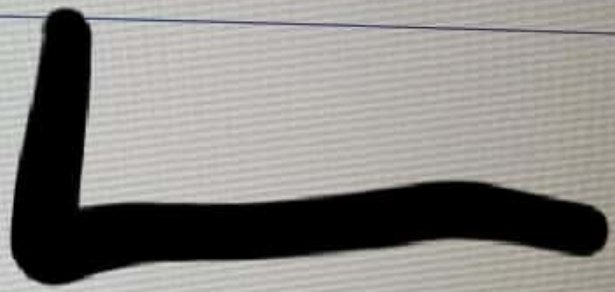
MTH631:Grand Quiz

Question # 1 of 30 (Start time: 08:28:37 AM, 12 July 2020)

Which of the following can be served as counterexample for scalar multiplication operation in $\mathbb{R}^n$ for the axiom: $1X = X, \forall X \in \mathbb{R}^n$?

Select the correct option

- $\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n), \alpha \in \mathbb{R}$
- $\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, 0, \ldots, 0), \alpha \in \mathbb{R}$



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Question # 2 of 30 (Start time: 08:32:43 AM, 12 July 2020)

Quiz Start Time: 08:31 AM,

Which of the followings does not hold for an inner product function $\langle .,. \rangle : V \times V \to F$.

Select the correct option

○	$\langle v, u \rangle = \langle u, v \rangle$
○	$\langle u + v, w \rangle = \langle v, w \rangle + \langle u, w \rangle$
○	$\langle cu, v \rangle = c \langle u, cv \rangle$
○	Non of these

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MTH631:Grand Quiz

Time Left 86 sec(s)

Quiz Start Time: 08:31 AM, 12 July 2020

Question # 1 of 30 (Start time: 08:31:51 AM, 12 July 2020)

Total Marks: 1

If the sequence of values $\{F_n(x)\}$ converges for each x in some subset S of D then defines a _____ on S

Select the correct option

Reload Math Equations

- limit function
- continuous function
- constant function
- function

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